

# ASSEM RYAN

Mechanical Engineer  
Manufacturing, Design & Test Automation



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## SUMMARY

Mechanical engineer and master's student with practical experience in CNC manufacturing, machine maintenance, CAD/CAM and test-bench automation. My profile combines manufacturing expertise with mechanical design, data acquisition and technical troubleshooting. I work effectively at the interface of production, product development and test engineering.

## TECHNICAL COMPETENCIES

- CNC manufacturing, process optimization and preventive maintenance
- Design for manufacturing of mechanical parts and assemblies
- CAD/CAM-based manufacturing preparation
- Test-bench automation and technical test procedures
- Data acquisition, signal visualization and system analysis
- Commissioning and parameterization of electromechanical systems
- Systematic troubleshooting, condition monitoring and root-cause analysis
- Basic FEM analysis, prototype development and additive manufacturing

## PROFESSIONAL EXPERIENCE

### Mechanical Engineer, Qallins Steel Nail Factory

Mar. 2022 — Jul. 2024

- Planned and coordinated preventive maintenance activities for production machinery
- Analyzed technical machine faults and identified potential root causes
- Implemented measures to reduce unplanned machine downtime
- Monitored machine condition and analyzed recurring technical issues
- Supported stable production processes in cooperation with the manufacturing team

### Mechanical Engineer, Samir's Workshop

Jun. 2021 — Mar. 2022

- Operated and monitored CNC turning machines for the production of mechanical components
- Monitored relevant machining parameters and inspected manufactured components
- Identified manufacturing deviations and supported technical troubleshooting
- Optimized machining procedures and technical workflows

## SOFTWARE SKILLS

- |              |                    |                           |                   |
|--------------|--------------------|---------------------------|-------------------|
| • SolidCAM   | • AutoCAD          | • MATLAB                  | • Cura            |
| • SolidWorks | • Ansys Mechanical | • Arduino IDE             | • Microsoft Excel |
| • Fusion 360 | • NI LabVIEW       | • SKF Monitoring Software |                   |

## SELECTED PROJECTS

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### PMSM Control Using Hybrid-Kit and One Eye

May 2025

*Dortmund University of Applied Sciences, Germany*

- Commissioning and parameterization of a permanent magnet synchronous motor
- Data acquisition and visualization using One Eye
- Analysis of system behavior and relevant operating parameters

### Test-Bench Automation – Nuevo-35 Automotive Electronics ↗

Apr. 2025 – Apr. 2026

- Optimization of a LabVIEW-based test bench with ramp-up and ramp-down functions
- Correction of errors in voltage and time calculations and adjustment of the control logic
- Implementation of a controlled shutdown with restart lockout to improve operational safety

## EDUCATION

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### Master's Degree in Mechanical Engineering

Apr. 2025 – Today

*Dortmund University of Applied Sciences, Germany*

- Specialization: Product Development and Simulation
- Master's project: Test-Bench Automation – Nuevo-35 Automotive Electronics
- Expected graduation: March 2027

### Bachelor's Degree in Production Engineering and Mechanical Design

Sept. 2015 – Sept. 2020

*Kafrelsheikh University, Egypt*

- Abschlussarbeit: Entwurf eines 3D-Druckers mit einem Lasergravierer ↗

## FURTHER EDUCATION

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Postgraduate Studies in Production Engineering and Mechanical Design    Feb. 2022 – Jul. 2023

*Tanta University, Egypt*

- Advanced coursework in robotics, production automation, mechanical design, and the optimization of modern manufacturing processes
- Overall grade: 2.0

## LANGUAGES

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- German: C1 – telc German C1 University ↗
- English: C1 – IELTS ↗
- Arabic: Native

## AWARDS & HONORS

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### 3rd Place – MittelstandsMakerthon NRW, 2026

- Development of an innovative concept in the field of surgical robotics
- Contribution to the technical concept development and presentation before an expert jury